



# Security Assessment



## Item 9 Internal Report

August 2026

Prepared for ether.fi



## Disclaimer

---

This document is intended for **internal** use only.



| Initial Commit Hash      | Final Commit Hash        |
|--------------------------|--------------------------|
| <a href="#">90e68688</a> | <a href="#">90e68688</a> |

## Protocol Scope

- src/enso/EnsoSwapModule.sol

## Informational Issues

---

### I-01. Keeper's native fee can be redirected by replacing the stored swap

**Description:** `executeSwap(address safe)` takes only the safe address and accepts any `nativeFee` equal to whatever `nativeFee` is stored at execution time.

The safe owner can therefore `cancelSwap` and immediately `requestSwapWithNativeFee` a cheap or no-op swap carrying the same signed fee. A keeper that quoted 0.01 ETH for a large legitimate swap then funds the replacement instead, and the surplus is refunded to an address named in the owner-supplied `swapData` (the standard bridge refund-address argument).

This is likely not an issue in practice because safe owners are only able to perform actions through the EtherFi UI without the ability to sign orders themselves through other means. On top of this, it is infeasible to perform front-running on OP so that the safe owner plants a new swap within the 10-second withdrawal delay period.

**Recommendation:** Add an expected-order parameter: `executeSwap(address safe, bytes32 expectedSwapId)`, reverting on mismatch.

**Customer's response:** Acknowledged.

# Disclaimer

Even though we hope this information is helpful, we provide no warranty of any kind, explicit or implied. The contents of this report should not be construed as a complete guarantee that the contract is secure in all dimensions. In no event shall Certora or any of its employees be liable for any claim, damages, or other liability, whether in an action of contract, tort, or otherwise, arising from, out of, or in connection with the results reported here.

# About Certora

Certora is a Web3 security company that provides industry-leading formal verification tools and smart contract audits. Certora's flagship security product, Certora Prover, is a unique SaaS product that automatically locates even the most rare & hard-to-find bugs on your smart contracts or mathematically proves their absence. The Certora Prover plugs into your standard deployment pipeline. It is helpful for smart contract developers and security researchers during auditing and bug bounties.

Certora also provides services such as auditing, formal verification projects, and incident response.